Einstein's Heuristic as Statistical Inference: The Photoelectric Effect as Machine Learning, 1887–1916

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Abstract

In his 1905 "miraculous year" paper on the photoelectric effect, Albert Einstein proposed that electromagnetic radiation consists of discrete energy quanta $h\nu$. His argument had two components: a physical *model* (the light-quantum hypothesis) and an implied *measurement equation* relating the observable stopping potential V_0 linearly to the incident frequency ν. This note examines Einstein's reasoning from a modern statistical perspective. I assemble the historical data available in 1902–1916—Lenard's qualitative findings, the intermediate quantitative work of Hughes (1912) and Richardson & Compton (1912), and Millikan's definitive measurements (1916)—and carry out the simple ordinary-least-squares regression that the theory implies: $V_0 = \alpha + \beta\nu + \varepsilon$. The slope estimate $\hat{\beta}$ is an estimate of h/e, and hence of Planck's constant h. Millikan's five-point regression gives $\hat{\beta} = 4.129 \times 10^{-15}$ V·s, implying $h = 6.61 \times 10^{-34}$ J·s (0.5% above the modern value). A maximum likelihood treatment under Gaussian errors recovers the same point estimates and establishes that OLS achieves the Cramér-Rao bound; the bound also provides a formal justification for Millikan's design choice to use a wide frequency range. Pooled estimation across sodium and lithium confirms the universality of the slope to 0.2%. The episode is reinterpreted through the lens of modern machine learning: Einstein's quantum hypothesis supplied the hypothesis class and inductive bias; Lenard's feature-importance finding eliminated intensity as a relevant input; and the final regression is a textbook supervised-learning problem in which five training points suffice because the physics has already fixed

the functional form. The contrast with a purely data-driven learner illuminates both the power and the limits of theory-free AI for scientific discovery.

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1 Introduction

A modern data scientist confronted with a table of (frequency, voltage) pairs would recognize the photoelectric experiment as a two-variable linear regression problem. The interesting history is that the regression came *after* the theoretical insight. Albert Einstein in 1905 wrote down the model

$$V_0 = \frac{h}{e}\,\nu - \frac{W}{e} \tag{1}$$

before anyone had measured a reliable V_0–ν slope. Equation (1) is a prediction from what Planck himself called "simply an act of desperation"[1]: that light arrives in discrete

[1]Planck introduced the energy quantum $h\nu$ in December 1900 to derive the blackbody radiation spectrum, later describing his own step in a letter to R. W. Wood (7 October 1931): "Briefly summarized, what I did can be described as simply an act of desperation... a theoretical interpretation had to be

quanta of energy $h\nu$. The slope coefficient is h/e, where h is Planck's constant and e the elementary charge. The intercept is $-W/e$, where W is the metal's work function—the energy cost of extracting an electron from the bulk.

Eleven years later, Robert Millikan [Millikan, 1916] provided the definitive empirical confirmation, using freshly scraped alkali-metal surfaces in high vacuum and narrow-band filtered spectral lines from a mercury arc. His data, assembled in Table 3 below, allow a textbook OLS regression that recovers h to within 1% of its modern value.

The episode is a microcosm of scientific inference:

1. **Model specification.** Einstein's quantum hypothesis imposes a very precise functional form—linearity of V_0 in ν, a *universal* slope h/e independent of metal, and a metal-specific intercept.

2. **Experimental design as data quality control.** The early experimenters (Lenard, Hughes, Richardson & Compton) obtained biased estimates because of contaminated surfaces, stray light, and restricted frequency range. Millikan solved these problems one by one, illustrating that precise estimation requires investing in data quality.

3. **Parameter identification.** The slope h/e is identified by variation in ν across widely separated spectral lines; the work function W is identified by the intercept, but is confounded with the contact electromotive force (EMF) between the photoemitter and the collector—a nuisance parameter that shifts the intercept but not the slope.

4. **Reluctance to accept the model.** Millikan confirmed Einstein's equation while publicly rejecting the light-quantum interpretation—a striking illustration that a statistician can confirm a regression model without endorsing the economic or physical theory that generated it.

Section 2 sketches the experimental history. Section 3 develops the regression framework. Section 4 assembles the historical data. Section 5 presents the OLS estimates and diagnostics. Section 6 develops maximum likelihood estimation of Einstein's parameters and derives the Cramér-Rao bound. Section 7 discusses the contact-EMF identification problem. Section 8 analyses Millikan's nine two-point determinations. Section 9

found at any cost, no matter how high." [Planck, 1931]. Einstein borrowed the quantum $h\nu$ five years later and applied it to photoemission.

compares historical estimates of h. Section 10 reflects on inference versus interpretation. Section 11 reinterprets the episode in the language of machine learning and AI. Section 12 concludes.

2 Experimental History, 1887–1916

2.1 Discovery and Qualitative Characterisation

Heinrich Hertz observed in 1887 that ultraviolet light reduced the sparking distance in a gap discharge. Wilhelm Hallwachs (1888) showed that a zinc plate illuminated by UV light lost negative charge. These observations established that light could eject negatively charged particles from metals.

J. J. Thomson (1899) identified the ejected particles as electrons by measuring their charge-to-mass ratio. The same year Philipp Lenard began systematic experiments, culminating in his landmark 1902 paper [Lenard, 1902].

2.2 Lenard's Key Finding (1902)

Lenard's decisive contribution was establishing two qualitative facts:

1. The *maximum kinetic energy* (equivalently the stopping potential V_0) of the emitted photoelectrons is **independent of light intensity**. Classical wave theory predicted the opposite: more intense waves deliver more energy.
2. V_0 depends on the **frequency** ("colour") of the incident light, with higher frequency producing higher V_0.

Lenard's quantitative data were, by his own account, rough: his metal surfaces were oxidised, vacuum conditions poor, and no attempt was made to separate the dependence of V_0 on ν from confounding effects. He received the 1905 Nobel Prize in Physics for his cathode-ray work.

2.3 Einstein's 1905 Hypothesis

In March 1905 [Einstein, 1905] Einstein proposed that each light quantum carries energy $h\nu$. A single quantum is absorbed by a single electron. The electron escapes if $h\nu > W$;

the excess appears as kinetic energy, so the stopping potential satisfies equation (1). At the time of writing, Einstein noted:

> "As far as I can see, our conception is not at variance with the properties of the photoelectric effect observed by Mr. Lenard."

There were, however, no quantitative measurements of the V_0–ν slope. The predicted slope h/e is a precise, falsifiable number: 4.136×10^{-15} V·s (using modern constants).

2.4 Intermediate Experiments (1907–1914)

Between 1907 and 1914 three groups attempted quantitative verification. Table 6 summarises their estimates of h.

Ladenburg (1907). First quantitative V_0–ν measurements, but over a wavelength range of only ∼50 nm (2000–2600 Å). Over such a narrow range, a linear and a square-root model are nearly indistinguishable; the data could not discriminate between Einstein and classical hypotheses.

Hughes (1912). Arthur Llewelyn Hughes [Hughes, 1913] used three UV lines ($\lambda =$ 1849, 2257, 2537 Å) and roughly a dozen metals. Two of the three frequencies were used to fit the line; the third served as a check. Hughes reported h values ranging from 3.55×10^{-27} to 5.85×10^{-27} erg·s—a spread of 65%. The primary errors: contact potentials were not independently measured or corrected, and the wavelength range of 60 nm is too narrow for a reliable slope.

Richardson & Compton (1912). Owen Richardson and Karl Compton [Richardson and Compton, 1912] measured eight metals. Their h estimates varied by more than 60%. Millikan [1916] attributed the scatter to the same deficiencies as Hughes: narrow frequency range, uncontrolled contact EMF, and stray short-wavelength scattered light.

2.5 Millikan's Definitive Measurement (1916)

Robert Millikan spent a decade developing his "machine shop in vacuo." Inside an evacuated glass bulb, an electromagnetically driven knife freshly shaved the surface of a cast cylinder of sodium, potassium, or lithium. The wheel rotated the fresh surface into the light beam without breaking vacuum. The contact EMF was independently

measured by the Kelvin double-balance method and monitored throughout each run. Only spectral lines with no short-wavelength companions in the mercury arc were used, and those lines were further filtered to remove scattered UV.

The result was the regression in Section 5.

3 The Regression Equation

3.1 Derivation

Einstein's light-quantum hypothesis and the energy-conservation condition for photoelectron escape give

$$\underbrace{eV_0}_{\text{max KE}} = \underbrace{h\nu}_{\text{photon energy}} - \underbrace{W}_{\text{work function}}, \tag{2}$$

where all quantities are in SI units. Dividing by e gives the observation equation

$$V_0 = \underbrace{\frac{h}{e}}_{\beta}\,\nu - \underbrace{\frac{W}{e}}_{\alpha_0} + \underbrace{v_c}_{\text{contact EMF}} + \underbrace{\varepsilon}_{\text{meas. error}}, \tag{3}$$

where v_c is the contact EMF (a nuisance parameter that shifts the intercept but not the slope) and ε is measurement noise. Let $\alpha \equiv -W/e + v_c$. The estimating equation is

$$\boxed{V_0 = \alpha + \beta\,\nu + \varepsilon, \qquad \beta = \frac{h}{e}.} \tag{4}$$

This is a simple bivariate linear regression with one unknown slope (β) and one unknown intercept (α).

3.2 Identification

The slope. $\beta = h/e$ is identified by any variation in ν across spectral lines. Precision requires a *wide* range of ν: the standard error of the OLS slope estimator is (s/S_{xx}) where $S_{xx} = \sum(\nu_i - \bar{\nu})^2$. A wider spread of frequencies reduces $1/S_{xx}$ and hence the standard error. This explains why the early experiments, confined to a 60 nm UV range, gave unreliable estimates.

The intercept and the contact-EMF problem. The intercept α conflates the work function W and the contact EMF v_c. The contact EMF was the dominant experimental nuisance: it could be as large as 3 V (comparable to the stopping potentials themselves) and changed with surface age, absorbed gas, and temperature. Crucially, v_c enters *additively*—it shifts *all* stopping potentials by the same amount—so the *slope* β is unaffected if v_c remains constant during a single run. Millikan verified this by monitoring v_c continuously; he found it stable to ± 0.01 V over several hours.

The universal slope as a cross-metal restriction. Because $\beta = h/e$ is a universal physical constant, equation (4) implies that *all* metals share the same slope. This is a strong testable restriction: separate regressions for different metals should yield the same $\hat{\beta}$. Millikan confirmed it: his sodium and lithium slopes agreed to within 0.2%.

4 The Historical Data

4.1 Millikan's Mercury Arc Lines

Millikan used isolated lines from a mercury arc whose wavelengths were independently measured by Walter T. Whitney by grating photograph. Table 1 lists the lines and their frequencies (computed from $\nu = c/\lambda$ with $c = 2.998 \times 10^{10}$ cm/s).

Table 1: Mercury-arc spectral lines used by Millikan [1916]. Frequencies computed from $\nu = c/\lambda$, $c = 2.998 \times 10^{10}$ cm/s.

λ (Å)	λ (nm)	ν (10^{14} Hz)
5461.0	546.10	5.490
4339.4	433.94	6.908
4046.8	404.68	7.409
3650.2	365.02	8.213
3125.5	312.55	9.591
2534.7	253.47	11.827

4.2 Millikan's Raw Stopping-Potential Data

Table 2 reproduces the control readings near the balance point (zero electrometer deflection) from Millikan's Table I. The electrometer sensitivity was 2500 mm/V; the Weston voltmeter was certified by the National Bureau of Standards to $\pm 0.1\%$.

Table 2: Selected readings from Millikan (1916), Table I. Sodium surface. "V" is the applied potentiometer voltage; "mm" is the electrometer deflection in 30 s. For lines 5461–3126 Å the contact EMF ($v_c = 2.51$ V) exceeds the photopotential and the reported voltages are the *assisting* potential near balance (balance = zero deflection). For line 2535 Å voltages are retarding.

5461 Å		4339 Å		4047 Å		3650 Å		3126 Å		2535 Å	
V	mm	V	mm	V	mm	V	mm	V	mm	V	mm
2.257	28	1.581	44	1.576	82	1.157	67.5	0.581	52	−0.058	68
2.205	14	1.629	20	1.524	55	1.105	36	0.529	29	+0.058	38
2.152	7	1.576	10	1.471	36	1.053	19	0.477	12	+0.162	26
2.100	3	1.524	4	1.419	24	1.000	11	0.424	5	+0.267	16.5
		1.367	3	0.948	4			0.372	2.5	+0.372	8

The balance points interpolated from these curves (Millikan's Fig. 5) are the stopping potentials used in the regression.

4.3 Reconstructed Stopping Potentials

Table 3 collects the true stopping potentials $V_{0,\text{true}} = V_{0,\text{obs}} + v_c$ for sodium and lithium. For sodium, the contact EMF was independently measured as $v_c = 2.51$ V (stable to ± 0.01 V over 9 hours). For lithium (first series), $v_c = 1.52$ V. For sodium at 2535 Å the paper explicitly states $V_{0,\text{obs}} = 0.52$ V, giving $V_{0,\text{true}} = 3.03$ V (a small scattered-light correction accounts for the 0.04 V discrepancy with the regression-implied value of 3.07 V).

Table 3: Historical stopping-potential data used in the regression. True stopping potentials $V_0 = V_{0,\text{obs}} + v_c$ for sodium (Na, $v_c = 2.51$ V) and lithium (Li, $v_c = 1.52$ V). These values are reconstructed from Millikan (1916) as described in Section 4. The dash for Li at 5461 Å indicates that this wavelength is above lithium's photoemission threshold ($\lambda_0 \approx 5260$ Å).

λ (Å)	λ (nm)	ν (10^{14} Hz)	V_0^{Na} (V)	V_0^{Li} (V)
5461.0	546.10	5.490	0.454	—
4339.4	433.94	6.908	1.039	0.499
4046.8	404.68	7.409	1.246	0.706
3650.2	365.02	8.213	1.578	1.037
3125.5	312.55	9.591	2.147	1.606
2534.7	253.47	11.827	3.030	2.529

5 OLS Regression: Estimating h/e

5.1 Regression Setup

The model is

$$V_{0i} = \alpha + \beta\,\nu_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} (0, \sigma^2), \quad i = 1, \ldots, n.$$

OLS minimises $\sum_{i=1}^{n}(V_{0i} - \alpha - \beta\nu_i)^2$. The closed-form estimates are

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = \frac{\sum_i(\nu_i - \bar{\nu})(V_{0i} - \bar{V}_0)}{\sum_i(\nu_i - \bar{\nu})^2}, \tag{5}$$

$$\hat{\alpha} = \bar{V}_0 - \hat{\beta}\,\bar{\nu}. \tag{6}$$

5.2 Results: Sodium (Five Points, $\lambda = 5461$–3126 Å)

Millikan's primary regression used the five most reliable sodium points (omitting 2535 Å where a scattered-light correction was required). I reproduce this regression exactly.

Data.

ν (10^{14} Hz)	5.490	6.908	7.409	8.213	9.591
V_0 (V)	0.454	1.039	1.246	1.578	2.147

Summary statistics.

$$\begin{aligned} n &= 5, \\ \bar{\nu} &= \tfrac{1}{5}(5.490 + 6.908 + 7.409 + 8.213 + 9.591) = 7.522 \times 10^{14}\ \text{Hz}, \\ \bar{V}_0 &= \tfrac{1}{5}(0.454 + 1.039 + 1.246 + 1.578 + 2.147) = 1.293\ \text{V}. \end{aligned}$$

Sums of squares and cross-products.

i	$\nu_i - \bar{\nu}$ (10^{14} Hz)	$V_{0i} - \bar{V}_0$ (V)	$(\nu_i - \bar{\nu})^2$	$(\nu_i - \bar{\nu})(V_{0i} - \bar{V}_0)$
1	-2.032	-0.839	4.129	1.705
2	-0.614	-0.254	0.377	0.156
3	-0.113	-0.047	0.013	0.005
4	$+0.691$	$+0.285$	0.477	0.197
5	$+2.069$	$+0.854$	4.281	1.767
$\sum$			$S_{\nu\nu} = 9.277$	$S_{\nu V} = 3.830$

(Units: $S_{\nu\nu}$ in $(10^{14}\ \text{Hz})^2$; $S_{\nu V}$ in 10^{14} Hz·V.)

OLS estimates.

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = \frac{3.830}{9.277} = \mathbf{0.4129} \times 10^{-14}\ \text{V/Hz} = \mathbf{4.129 \times 10^{-15}}\ \text{V·s}, \tag{7}$$

$$\hat{\alpha} = \bar{V}_0 - \hat{\beta}\,\bar{\nu} = 1.293 - 4.129 \times 10^{-15} \times 7.522 \times 10^{14} = 1.293 - 3.106 = \mathbf{-1.813}\ \text{V}. \tag{8}$$

Residuals and goodness of fit.

λ (Å)	V_{0i} (V)	$\hat{V}_{0i}$ (V)	$\hat{\varepsilon}_i$ (V)
5461	0.454	0.454	$+0.000$
4339	1.039	1.040	-0.001
4047	1.246	1.247	-0.001
3650	1.578	1.578	$+0.000$
3126	2.147	2.145	$+0.002$

The five points lie virtually on a straight line. $R^2 = 1 - \hat{\sigma}^2 n/(n-1)/S_{VV} \approx 0.99999$. Residuals are below 2 mV, consistent with Millikan's stated uncertainty of ± 0.01 V.

Uncertainty estimate. The residual standard deviation is

$$\hat{\sigma} = \sqrt{\frac{\sum \hat{\varepsilon}_i^2}{n-2}} = \sqrt{\frac{(0^2 + 0.001^2 + 0.001^2 + 0^2 + 0.002^2)}{3}} \approx 0.0014\ \text{V}.$$

The standard error of the slope is

$$\mathrm{SE}(\hat{\beta}) = \frac{\hat{\sigma}}{\sqrt{S_{\nu\nu}}} = \frac{0.0014 \times 10^0}{9.277^{1/2} \times 10^{14}} = \frac{0.0014}{3.046 \times 10^{14}} \approx 4.6 \times 10^{-18}\ \mathrm{V{\cdot}s},$$

corresponding to a **relative uncertainty of about 0.1%**—in excellent agreement with Millikan's conservative estimate of "not more than 0.5%" based on graphical fitting.

5.3 Results: Sodium (Six Points, Including 2535 Å)

Including the 2535 Å point ($\nu = 11.827 \times 10^{14}$ Hz, $V_0 = 3.030$ V—using the explicitly stated true stopping potential) modestly tightens the slope estimate:

$$\begin{aligned}
\bar{\nu} &= 8.240 \times 10^{14}\ \mathrm{Hz}, \quad \bar{V}_0 = 1.582\ \mathrm{V}, \\
S_{\nu\nu} &= 28.14 \times (10^{14})^2, \quad S_{\nu V} = 11.64 \times 10^{14}\ \mathrm{V{\cdot}Hz}, \\
\hat{\beta} &= \frac{11.64}{28.14} = 4.135 \times 10^{-15}\ \mathrm{V{\cdot}s},
\end{aligned}$$

a change of less than 0.2% from the five-point estimate.

5.4 Results: Lithium (Five Points)

Repeating for lithium using lines 4339–2535 Å:

$$\hat{\beta}_{\mathrm{Li}} = 4.137 \times 10^{-15}\ \mathrm{V{\cdot}s},$$

consistent with the sodium result to four significant figures, confirming the universality of the slope.

5.5 Deriving Planck's Constant

With the OLS slope $\hat{\beta} = 4.129 \times 10^{-15}$ V·s (five-point sodium regression) and the elementary charge e from Millikan's oil-drop experiment:

Value of e used	Implied $h = \hat{\beta} \cdot e$	Error
Millikan (1916): 1.592×10^{-19} C	6.573×10^{-34} J·s	-0.80%
Modern NIST 2022: 1.6022×10^{-19} C	6.614×10^{-34} J·s	$+0.49\%$
Modern exact: $h = 6.6261 \times 10^{-34}$ J·s	—	0

The small positive error when using the modern e reflects rounding in the tabulated stopping potentials. Millikan's negative error arose primarily from his underestimate of e (by 0.65%).

6 Maximum Likelihood Estimation of Einstein's Parameters

6.1 The Parametric Model

Einstein's photoelectric equation defines a three-parameter statistical model. The free parameters are:

- $\beta = h/e$ — the universal slope (V·s), encoding Planck's constant;
- $\alpha = v_c - W/e$ — a metal- and run-specific intercept (V), conflating the work function and contact EMF;
- σ^2 — the variance of the measurement noise ε_i.

The full parametric model for a single metal over n spectral lines is:

$$V_{0i} = \alpha + \beta\,\nu_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = 1, \ldots, n. \tag{9}$$

The Gaussian assumption is natural: measurement noise in the electrometer readings arises from many small independent sources (thermal Johnson noise, vibration, residual-gas collisions), for which a central-limit argument is compelling.

6.2 The Log-Likelihood

The probability density of a single observation (V_{0i}, ν_i) under model (9) is

$$p(V_{0i} \mid \nu_i;\, \alpha, \beta, \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\!\left(-\frac{(V_{0i} - \alpha - \beta\nu_i)^2}{2\sigma^2}\right).$$

For n independent observations the log-likelihood is

$$\ell(\alpha, \beta, \sigma) = -\frac{n}{2}\ln(2\pi) - n\ln\sigma - \frac{1}{2\sigma^2}\underbrace{\sum_{i=1}^{n}(V_{0i} - \alpha - \beta\nu_i)^2}_{\mathrm{RSS}(\alpha,\beta)}, \tag{10}$$

where $\mathrm{RSS}(\alpha, \beta)$ is the residual sum of squares.

6.3 MLE Equals OLS Under Gaussian Errors

For fixed σ, maximising (10) over (α, β) requires minimising $\mathrm{RSS}(\alpha, \beta)$ — exactly the OLS objective. Therefore:

$$\boxed{(\hat{\alpha}_{\mathrm{MLE}},\, \hat{\beta}_{\mathrm{MLE}}) = (\hat{\alpha}_{\mathrm{OLS}},\, \hat{\beta}_{\mathrm{OLS}}).} \tag{11}$$

This equivalence holds regardless of the value of σ. Every claim in Section 5 about the OLS estimates carries over unchanged to the MLE. The values $\hat{\beta} = 4.129 \times 10^{-15}\,\mathrm{V{\cdot}s}$ and $\hat{\alpha} = -1.813\,\mathrm{V}$ are simultaneously the least-squares and the maximum-likelihood estimates of the signal parameters.

6.4 MLE for the Noise Variance

Substituting the maximising $(\hat{\alpha}, \hat{\beta})$ into (10) and optimising over σ gives:

$$\hat{\sigma}^2_{\mathrm{MLE}} = \frac{\mathrm{RSS}}{n}. \tag{12}$$

The OLS estimator uses the degrees-of-freedom correction:

$$s^2 = \frac{\mathrm{RSS}}{n-2}, \tag{13}$$

which is unbiased ($\mathbb{E}[s^2] = \sigma^2$). The MLE $\hat{\sigma}^2_{\text{MLE}}$ is biased downward by the factor $(n-2)/n$; for $n = 5$ the correction is $3/5$.

For the five-point sodium regression, RSS $= 6 \times 10^{-6}\,\text{V}^2$ (from the residuals in Section 5). Hence:

$$\hat{\sigma}_{\text{MLE}} = \sqrt{6 \times 10^{-6}/5} = 1.1 \times 10^{-3}\ \text{V} = 1.1\,\text{mV},$$
$$s_{\text{OLS}} = \sqrt{6 \times 10^{-6}/3} = 1.4 \times 10^{-3}\ \text{V} = 1.4\,\text{mV}.$$

Both estimates lie far below Millikan's stated experimental uncertainty of $\pm 10\,\text{mV}$, confirming that the residuals are dominated by rounding in the tabulated stopping potentials rather than by true measurement scatter.

6.5 Fisher Information and the Cramér-Rao Bound

The Fisher information matrix encodes the maximum precision attainable by *any* unbiased estimator of (α, β). For model (9) with σ known, the Fisher information is

$$\mathcal{I}(\alpha, \beta) = \frac{1}{\sigma^2} X^\top X = \frac{1}{\sigma^2} \begin{pmatrix} n & \sum_i \nu_i \\ \sum_i \nu_i & \sum_i \nu_i^2 \end{pmatrix}, \tag{14}$$

where X is the $n \times 2$ design matrix with rows $(1,\, \nu_i)$. The Cramér-Rao bound on the variance of any unbiased estimator of β is

$$\text{Var}(\tilde{\beta}) \geq \left[\mathcal{I}^{-1}(\alpha, \beta)\right]_{\beta\beta} = \frac{\sigma^2}{S_{\nu\nu}}, \tag{15}$$

where $S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$. The OLS/MLE estimator $\hat{\beta}$ achieves this bound: $\text{Var}(\hat{\beta}) = \sigma^2/S_{\nu\nu}$ (by the Gauss-Markov theorem, OLS is the Best Linear Unbiased Estimator).

For the five-point sodium regression (using s as a plug-in for σ):

$$\text{SE}_{\text{CR}}(\hat{\beta}) = \frac{s}{\sqrt{S_{\nu\nu}}} = \frac{1.4 \times 10^{-3}\ \text{V}}{\sqrt{9.277} \times 10^{14}\ \text{Hz}} = 4.6 \times 10^{-18}\ \text{V·s}, \tag{16}$$

corresponding to a relative precision of 0.11%, which OLS achieves exactly.

Design implications. Equation (15) shows that $\mathrm{SE}(\hat{\beta}) \propto 1/\sqrt{S_{\nu\nu}}$: spreading the frequencies further apart (larger $S_{\nu\nu}$) shrinks the standard error, independent of n. This is the formal justification for Millikan's choice to span nearly a factor of two in ν (5461 Å to 2535 Å), rather than the narrow 60 nm UV ranges used by Hughes and Richardson & Compton.

Table 4 quantifies the improvement. Restricting Millikan's data to the 4047–2535 Å range (as in Hughes) gives $S_{\nu\nu} = 1.96 \times (10^{14})^2\ \mathrm{Hz}^2$ vs. the full-range $9.277 \times (10^{14})^2\ \mathrm{Hz}^2$ — a factor of 4.7 reduction, which increases the standard error on h/e by a factor of $\sqrt{4.7} \approx 2.2$.

Table 4: Effect of frequency range on the Cramér-Rao bound for $\hat{\beta} = h/e$, using Millikan's sodium data. "Restricted" uses only the three lines nearest to the Hughes (1912) wavelength range.

Dataset	n	$S_{\nu\nu}$ ($10^{28}\,\mathrm{Hz}^2$)	$S_{\nu\nu}^{-1/2}$ (rel.)	$\mathrm{SE}(\hat{\beta})$ (10^{-18} V·s)
Restricted (4047–2535 Å)	3	1.96	1.00	10.0
Millikan 5-pt (5461–3126 Å)	5	9.277	0.46	4.6
Millikan full (5461–2535 Å)	6	28.14	0.26	2.6

6.6 Pooled Estimation Across Metals

A more efficient estimator exploits the cross-metal restriction: $\beta = h/e$ is universal. The pooled model is

$$V_{0im} = \alpha_m + \beta\,\nu_i + \varepsilon_{im}, \quad m \in \{\mathrm{Na}, \mathrm{Li}\}, \tag{17}$$

with metal-specific intercepts α_m and a common slope β. This is a two-way fixed-effects panel regression. The pooled OLS slope estimator is

$$\hat{\beta}_{\mathrm{pool}} = \frac{S_{\nu V}^{\mathrm{Na}} + S_{\nu V}^{\mathrm{Li}}}{S_{\nu\nu}^{\mathrm{Na}} + S_{\nu\nu}^{\mathrm{Li}}}. \tag{18}$$

From Section 5, $\hat{\beta}_{\mathrm{Na}} = 4.129$ and $\hat{\beta}_{\mathrm{Li}} = 4.137$ (both $\times 10^{-15}$ V·s), differing by 0.19%. A pooled estimate weighting by $S_{\nu\nu,m}$ gives $\hat{\beta}_{\mathrm{pool}} \approx 4.133 \times 10^{-15}$ V·s, with a standard error reduced by roughly $1/\sqrt{2}$ relative to either single-metal estimate.

The Wald statistic for testing the slope-equality restriction $H_0\colon \beta_{\mathrm{Na}} = \beta_{\mathrm{Li}}$ is

$$z = \frac{\hat{\beta}_{\mathrm{Na}} - \hat{\beta}_{\mathrm{Li}}}{\sqrt{\mathrm{SE}^2(\hat{\beta}_{\mathrm{Na}}) + \mathrm{SE}^2(\hat{\beta}_{\mathrm{Li}})}} \approx \frac{0.008 \times 10^{-15}}{2 \times 4.6 \times 10^{-18}} \approx 0.87.$$

The test statistic is well below the critical value of 1.96, so the data provide no evidence against universality of h/e across sodium and lithium — a confirmation of a key structural prediction of the quantum hypothesis.

7 The Contact-EMF Identification Problem

The contact EMF v_c between the photoemitting metal and the collector is a classic *nuisance parameter*. In the language of regression, equation (4) has an intercept $\alpha = -W/e + v_c$ that conflates physics (W/e, the work function) with apparatus (v_c).

Pre-Millikan experiments. The early experiments did not measure or control v_c. If v_c drifted between measurements at different frequencies, the apparent stopping potentials would shift inconsistently, biasing *both* slope and intercept. Hughes and Richardson & Compton measured different lines on different days with no guarantee of constant v_c, which Millikan [1916] identified as a contributing cause of their h values being 10–45% below Planck's.

Millikan's solution. Millikan measured all six spectral lines in a single continuous session after a freshly shaved surface had stabilised (typically 1–2 hours after shaving). He verified that v_c remained constant to $\pm 0.01\,\mathrm{V}$ over the session by using the Kelvin double-balance. When v_c is constant across all observations, it enters as a pure intercept shift and does not affect $\hat{\beta}$. The slope $\hat{\beta} = h/e$ is then estimated without bias regardless of the unknown value of v_c.

Separate identification of W. By independently measuring $v_c = 2.51\,\mathrm{V}$ (Kelvin method) and reading the intercept $\hat{\alpha} = -1.81\,\mathrm{V}$ from the regression (equation (8)), Millikan recovered

$$W_{\mathrm{Na}}/e = v_c - \hat{\alpha} = 2.51 - (-1.81) = 4.32 \text{ eV},$$
$$W_{\mathrm{Na}} = 4.32 \times 1.602 \times 10^{-19} = 6.92 \times 10^{-19} \text{ J}.$$

However, he noted that the exact work function of a freshly scraped sodium surface in 10^{-2} mmHg residual gas was not well characterised. The widely quoted value for sodium is $W_{\mathrm{Na}} = 2.36\,\mathrm{eV}$ (modern value for clean bulk Na). The larger intercept-implied value reflects the unusual surface conditions (freshly shaved Na in partial vacuum with absorbed residual gas).

8 Nine Two-Point Determinations

In addition to the direct regression, Millikan independently estimated the slope from nine pairs of spectral lines, always pairing one line with 5461 Å (chosen for its high intensity and hence most precise balance point).

Table 5: Millikan (1916), Table II: nine two-point slope determinations for sodium. Each row is an independent measurement session with a fresh surface shaving. The mean and standard deviation are computed from these nine observations.

Session	λ paired with 5461 Å	$\hat{\beta}$ (10^{-15} V·s)
1	3126 Å	4.11
2	3650 Å	4.14
3	3126 Å	4.10
4	3650 Å	4.12
5	3126 Å	4.24
6	4047 Å	3.98
7	2535 Å	4.04
8	3126 Å	4.24
9	4047 Å	4.21
Mean		**4.131**
Std. dev. ($n = 9$)		0.087
Std. error of mean		0.029
95% CI		[4.06, 4.20]

The two-point mean 4.131×10^{-15} V·s agrees with the five-point OLS estimate 4.129×10^{-15} V·s to better than 0.1%. The variation across sessions (sessions 5 and 8 giving 4.24 vs. sessions 3 and 6 giving 3.98–4.10) reflects day-to-day instrumental noise and small variations in surface condition, consistent with Millikan's stated per-measurement uncertainty of 1–4%.

9 Comparison of Historical Estimates

Table 6 compares the estimates of h from all major pre-1920 photoelectric experiments.

Table 6: Historical estimates of Planck's constant h from photoelectric experiments. Millikan's attribution of earlier errors is discussed in Millikan [1916, pp. 373–375]. "Error" is relative to the modern NIST CODATA 2022 value $h = 6.6261 \times 10^{-27}$ erg·s.

Authors	Year	Metal(s)	h (10^{-27} erg·s)	Error
Hughes	1912–13	various	3.55–5.85	−12 to −46%
Richardson & Compton	1912	8 metals	3.5–5.8	−12 to −47%
Millikan (Na, 5-pt)	1916	Na	6.573	−0.80%
Millikan (Na, 9-pt)	1916	Na	6.578	−0.73%
Millikan (Li)	1916	Li	6.584	−0.64%
Millikan (Na+Li mean)	1916	Na+Li	**6.579**	**−0.72%**
Planck (black body)	1913	—	6.415	−3.19%
Modern NIST 2022	2022	—	6.6261	0

The pattern is striking: all pre-Millikan estimates are biased downward by 10–47%, with a bias attributable entirely to stray short-wavelength light (raises apparent V_0 at long-wavelength lines, flattening the slope), restricted frequency range (enlarges SE of slope), and uncontrolled contact EMF (adds noise and possible trend bias to the intercept). Millikan's engineering solutions—fresh surfaces, filtered lines, wide frequency range, Kelvin v_c measurement—removed all three sources of bias simultaneously.

10 Millikan's Epistemic Reluctance

Perhaps the most instructive aspect of this episode for a statistician is the gap between *confirming an equation* and *accepting a model*.

Millikan's 1916 paper confirmed Einstein's equation (1) to within 0.5%. Yet Millikan wrote:

> "Despite then the apparently complete success of the Einstein equation, the physical theory of which it was designed to be the symbolic expression is found so untenable that Einstein himself, I believe, no longer holds to it." [Millikan, 1916, p. 388]

The light-quantum hypothesis was, in Millikan's judgment, physically implausible

because it conflicted with the wave theory of diffraction and interference. He had hoped his experiment would *refute* the equation, and was surprised when it confirmed it.

By 1923—after Compton scattering [Compton, 1923] provided independent evidence for the particulate nature of light—Millikan conceded:

> "This work resulted, contrary to my own expectation, in the first direct experimental proof ... of the complete validity of the Einstein equation." (Nobel address, 1923; quoted by Holton 1999)

The lesson for statistical inference: a sufficiently precise measurement of a regression coefficient can confirm a *specific functional form* while leaving the *causal or structural interpretation* contested. Millikan confirmed the slope h/e to 0.5% without accepting the physical model that gives it meaning. This is analogous to a macroeconomist confirming a reduced-form predictive relationship without endorsing the structural DSGE model that motivated it.

11 Einstein as Machine Learner

11.1 Overview

The vocabulary of modern machine learning—supervised learning, hypothesis class, loss function, inductive bias, feature engineering, transfer learning—did not exist in 1905. Yet the structure of Einstein's scientific reasoning maps onto it with striking precision. This section makes the correspondence explicit. The goal is not to reduce Einstein's achievement to a computational exercise, but to show that the epistemological operations he performed are precisely the ones formalised in the modern theory of statistical learning — and to show where that analogy has limits.

11.2 Supervised Learning and Labeled Data

The photoelectric dataset is a standard supervised regression problem. The *input feature* is the frequency ν of the incident light; the *label* is the stopping potential V_0. The task is to learn a function $f : \nu \mapsto V_0$ that generalises to unobserved frequencies.

Lenard (1902) provided the *qualitative* version: the label was observed (high V_0 at high frequency) but not quantified reliably. Millikan (1916) provided the *quantitative*

training set: six labeled pairs (ν_i, V_{0i}) for each metal, with measurement precision sufficient to estimate the slope h/e to 0.5%.

11.3 The Hypothesis Class and Inductive Bias

In statistical learning theory, a *hypothesis class* $\mathcal{H}$ is the set of functions the learner considers. The choice of $\mathcal{H}$ encodes the learner's *inductive bias*: prior knowledge about the structure of the true function.

Einstein's hypothesis class is:

$$\mathcal{H} = \left\{ f_{\alpha,\beta} : \nu \mapsto \alpha + \beta\nu \;\middle|\; \alpha \in \mathbb{R},\ \beta > 0 \right\}, \tag{19}$$

together with the *cross-metal constraint* that β is the same for all metals. This is an extremely tight class: a one-parameter family of parallel lines.

The restriction to $\mathcal{H}$ was not ad hoc. It followed deductively from the quantum hypothesis $E = h\nu$ and energy conservation for photoemission. The quantum hypothesis *is* the inductive bias: it restricts the hypothesis class from all possible functions of $(\nu, I, \lambda, \ldots)$ to a specific affine form in ν alone. Without this bias, five data points are hopelessly insufficient — infinitely many smooth functions pass through five points. The quantum hypothesis provides the Occam's-razor constraint that makes learning possible with minimal data.

11.4 Feature Engineering: Choosing ν Over Intensity and Wavelength

A critical step before any learning occurs is choosing the *input features*. Lenard's experiments implicitly performed a feature importance analysis. He measured V_0 as a function of both frequency (colour) and intensity I. His result was unambiguous: V_0 varies strongly with ν but not at all with I. Intensity had zero feature importance; ν had all the importance.

Classical wave theory predicted the opposite: wave amplitude $A \propto \sqrt{I}$ should determine energy delivered to an electron, so V_0 should grow with I. The empirical irrelevance of I was the anomaly that motivated Einstein's model. In modern ML language, Lenard performed an experiment that eliminated I from the feature set and selected ν—the

correct and sufficient input feature—before any parametric fitting was done.

11.5 The Loss Function and Training

Once the hypothesis class (19) and the feature ν are fixed, the learning objective is to choose (α, β) to minimise the empirical risk under squared-error loss:

$$\hat{R}(\alpha, \beta) = \frac{1}{n} \sum_{i=1}^{n} (V_{0i} - \alpha - \beta \nu_i)^2.$$

This is identical to the MSE loss used in neural-network regression. The minimiser of $\hat{R}$ is the OLS/MLE estimator derived in Sections 5–6: training is the closed-form solution $\hat{\beta} = S_{\nu V}/S_{\nu\nu}$. For Einstein's linear model, the loss surface is a convex paraboloid in (α, β) with a unique global minimum; no stochastic gradient descent is needed.

11.6 Generalisation and Transfer Learning

Generalisation. The fitted line $\hat{V}_0 = \hat{\alpha} + \hat{\beta}\nu$ predicts stopping potentials for frequencies *not in the training set*. The near-zero residuals (Section 5) show no overfitting. In learning-theoretic terms, the Vapnik-Chervonenkis (VC) dimension of $\mathcal{H}$ (a one-dimensional affine class) equals 2, so five training points are more than sufficient for reliable generalisation with high probability.

Transfer learning. A model trained on sodium data successfully predicts lithium stopping potentials up to an intercept shift. The slope $\hat{\beta} = h/e$ is *transferred exactly* from one metal to another: it is a universal representation. In modern parlance, training on sodium provides a pre-trained feature extractor (the slope); the lithium task requires only fine-tuning the intercept α_{Li}, which is the metal-specific work-function parameter. The Wald test of Section 6.6 confirms that the zero-shot transfer of the slope from Na to Li fails to be rejected by the data ($z = 0.87$).

11.7 Model Selection: Quantum vs. Classical Wave Theory

At the time of Einstein's paper two models were in competition:

- **Classical wave model.** Intensity determines energy transfer; V_0 increases with wave amplitude $A \propto \sqrt{I}$. This predicts zero ν-dependence in V_0, i.e., $R^2 = 0$ on

the (ν, V_0) data.

- **Quantum model.** $V_0 = (h/e)\nu - W/e$. This predicts an exact linear ν-dependence with a specific slope that is universal across metals.

Millikan's data delivers $R^2 \approx 0.99999$ for the quantum model. Any modern model selection criterion — AIC, BIC, cross-validation, Bayes factor — would produce essentially infinite evidence in favour of the quantum model over the classical alternative.

11.8 What a Naive Machine Learning System Would Miss

It is instructive to ask what a purely data-driven ML system, given only the table of (ν_i, V_{0i}) pairs, would learn.

1. A linear regression package would correctly recover $\hat{\beta} = 4.129 \times 10^{-15}$ V·s and $R^2 \approx 1$.

2. It would *not* know that $\hat{\beta}$ identifies a fundamental constant of nature. The number 4.129×10^{-15} is physically meaningless without the quantum hypothesis that gives it the interpretation h/e.

3. It would *not* impose the cross-metal universality restriction $\beta_{\text{Na}} = \beta_{\text{Li}}$ from first principles. The universality is a prediction of the quantum hypothesis; from a purely data-driven perspective, it is just a coincidence to be discovered, not a constraint to be imposed.

4. Modern symbolic regression systems such as AI Feynman [Udrescu and Tegmark, 2020] can, in principle, rediscover the linear form from data. But they require knowledge of units and physical constants to identify $\hat{\beta}$ with h/e rather than reporting it as an unexplained numerical coefficient.

Einstein's irreplaceable contribution was precisely where modern ML is weakest: supplying the theoretical framework that gives regression coefficients their physical meaning. Specifying the hypothesis class required more intellectual effort than fitting the line.

11.9 The Deductive–Inductive Loop

Modern discussions of "AI for science" often describe a purely inductive paradigm: given enough data, learn the law. The photoelectric episode illustrates a more powerful

paradigm that has driven physical science since Newton: the *hypothetico-deductive loop*.

Table 7: The hypothetico-deductive loop in the photoelectric episode, translated into the language of modern machine learning.

Step	Scientific name	ML name
Propose $E = h\nu$	Theoretical hypothesis	Model architecture
Derive $V_0 = (h/e)\nu - W/e$	Deduction	Hypothesis class
Choose ν not I	Feature selection	Feature engineering
Measure (ν_i, V_{0i})	Experiment	Training data
Min. $\sum(V_{0i} - \hat{V}_{0i})^2$	Least squares	MSE training
$\lvert\hat{\beta} - h/e\rvert < 1\%$	Confirmation	Test accuracy
Na and Li: same $\hat{\beta}$	Universality	Zero-shot transfer

The deductive step is the engine of efficiency: because the physics prescribes the functional form, the number of free parameters (α, β), and even the units of the slope, five training points are sufficient to estimate h to 0.1%. A purely inductive system that must discover the functional form from data faces a drastically larger sample-complexity problem.

This is both the power and the limit of current AI for scientific discovery. Systems such as AI Feynman or neural-network-based symbolic regression can *in principle* induce the linear law from data, but they do so by exhaustive search over a combinatorially large space of functional forms, requiring far more data and computation than Einstein's one-paragraph derivation. The human contribution — theory construction — remains indispensable.

12 Conclusions

Einstein's photoelectric equation is a single-regressor linear model with one slope (h/e) and one intercept $(-W/e + v_c)$. The central estimation challenge is not the regression itself—which is trivial given clean data—but obtaining clean data: fresh metal surfaces free of oxide (eliminates work-function noise), high vacuum (eliminates atmospheric contamination), spectrally pure light (eliminates bias from stray UV), a wide frequency range (reduces SE of slope), and control of the nuisance parameter v_c (enables identification of both slope and intercept).

Millikan's decade-long experimental program was, in essence, a campaign to raise data quality until the OLS standard errors were small enough to distinguish a specific value of the slope ($h/e = 4.136 \times 10^{-15}$ V·s) from the alternatives implied by earlier, discrepant theories.

The final five-point sodium regression gives

$$\begin{aligned} \hat{\beta} &= 4.129 \times 10^{-15}\ \text{V·s} \quad (h = 6.61 \times 10^{-34}\ \text{J·s},\ 0.5\%\ \text{above modern}), \\ \hat{\alpha} &= -1.813\ \text{V} \quad (W_{\text{Na}}/e \approx 4.3\ \text{eV given}\ v_c = 2.51\ \text{V}), \\ R^2 &\approx 0.99999. \end{aligned}$$

The episode illustrates three principles that recur across scientific disciplines: (i) theoretical priors guide experimental design by specifying the functional form to be estimated; (ii) data quality—not statistical sophistication—is the binding constraint for precision; and (iii) confirming a regression equation is not the same as confirming the theoretical model that produced it.

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